DO NOW — RECALL

Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

1. Distinguish RAM from virtual storage.

2. In a BST, where is a value less than the current node placed?

3. Why can an array provide direct indexed access?

4. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

RAM is local working memory; virtual storage presents remote/pooled file storage.

2. Model answer

In its left subtree (under the chosen no-duplicate convention).

3. Model answer

Its element locations can be calculated from base address and index.

4. Model answer

Sum 0 and carry-out 1.

By the end, you can…

• The OS creates and schedules processes; programs share CPU time.

• Memory, files and device access are mediated by the OS.

• Interfaces, user accounts and access controls make a computer usable.

Vocabulary — know these terms

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

NEW LEARNING

Read, annotate and explain

Manage processes

The OS creates and schedules processes; programs share CPU time.

Manage resources

Memory, files and device access are mediated by the OS.

Provide services

Interfaces, user accounts and access controls make a computer usable.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A word processor saves a document while music is playing. Explain three OS services involved.

Model answer

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe three functions of an operating system, linking each to a user action. [6]

Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The operating system only provides a graphical user interface.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.